



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE(Main)

FULL TEST

29-12-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : TOAS, TNAS, TAAS, TEAS, TIAS, TMAS, I(A) & LIVE-I(A)] (SCORE-I)

Time : 3 Hours

PAPER-2

Maximum Marks : 300

READ THE INSTRUCTIONS CAREFULLY

Important Instructions :

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The Test Booklet consists of **90** questions.
4. There are **three** parts in the question paper 1,2,3 consisting of **Physics, Chemistry and Mathematics** having **30 questions** in each subject and each subject having **Two sections**.
 - (i) Section-I contains 20 multiple choice questions with **only one correct** option.
Marking scheme : +4 for correct answer, 0 if not attempted and -1 in all other cases.
 - (ii) Section-II contains 10 **Numerical Value Type** questions. Attempt any 5 questions. First 5 attempted questions will be considered for marking.
Marking scheme : +4 for correct answer, 0 if not attempted and -1 in all other cases.
5. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** and **Side-2** of the Answer Sheet. **Use of pencil is strictly prohibited.**
6. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
7. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
8. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
9. **Do not fold or make any stray marks on the Answer Sheet.**
10. Take $g = 10 \text{ m/s}^2$ unless otherwise stated.

DO NOT BREAK THESE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Name of the Candidate (in Capitals)_____

Form Number : in figures_____

: in words_____

Centre of Examination (in Capitals) : _____

Candidate's Signature : _____ Invigilator's Signature : _____

Your Target is to secure Good Rank in JEE(Main) 2023

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SECTION-I : (Maximum Marks: 80)

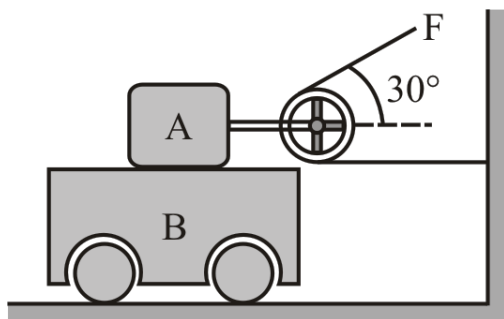
This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

Full Marks : +4 If correct answer is selected.

Zero Marks : 0 If none of the option is selected.

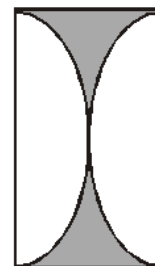
Negative Marks : -1 If wrong option is selected.

1. Block A of mass 1 kg is placed on a trolley B of mass 10 kg as shown. The coefficient of friction between A and B is 0.5, while the horizontal surface on which trolley is placed is smooth. If the value of F is half of the limiting value of F for no slipping between A and B, then the magnitude of friction force between A and B is :



- (A) 2.4N
(B) 2.57N
(C) 2.18N
(D) 1.285N

2. Two plano-convex lenses each of focal length 10 cm and refractive index $\frac{3}{2}$ are placed as shown. In the space left, water $\left(RI = \frac{4}{3}\right)$ is filled. The whole arrangement is in air. The optical power of the system is (in diopters)



- (A) 6.67
(B) -6.67
(C) 33.3
(D) 20
3. A current I increases at a steady rate in a long straight wire. The variation of current with time is $I = 3.36 (1 + 2t) \times 10^{-2}$ A. A small circular loop of radius 10^{-3} m has its plane parallel to the wire and is placed at a distance of 1 m from the wire. The resistance of the loop is $8.4 \times 10^{-4} \Omega$. The magnitude of the induced current in the loop is.
- (A) 1×10^{-11} A
(B) 3×10^{-11} A
(C) 5×10^{-11} A
(D) 7×10^{-11} A

4. Two coherent point sources, emitting light of wavelength λ , are placed at a separation of 4λ along positive y-axis. The number of minima, detected by a detector, which is travelling from $-\infty$ to $+\infty$ along x-axis, is

(A) 4
(B) 6
(C) 8
(D) 9

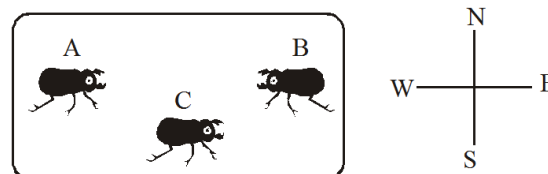
5. The engineer of a train blows the train whistle as he approaches a crossing. A few moments later he hears a echo from the whistle. The engineer hears the echo of the whistle because of:

(A) reflection
(B) refraction
(C) constructive interference
(D) destructive interference

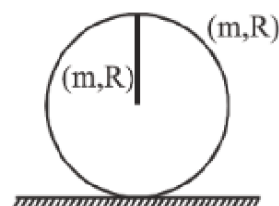
6. A TV tower has a height of 100 m. How much population is covered by TV broadcast, if the average population density around the tower is 1000 km^{-2} (radius of Earth = $6.4 \times 10^6 \text{ m}$) ?

(A) 10^3
(B) 10^6
(C) 4×10^6
(D) 4×10^9

7. Three bugs A, B and C of masses 3g, 5g and 2 g respectively are moving on a horizontal carboard placed on smooth table. Bug A is moving with an acceleration of 1 cm/s^2 in East direction, while bug C is moving with an acceleration of 2 cm/s^2 in North direction. Bug B is moving such that the cardboard remains stationary. Mark the correct option.



- (A) The bug B remains stationary
(B) The bug B moves with an acceleration 2.5 cm/s^2 in a direction 53° South of West.
(C) The bug B moves with an acceleration 1 cm/s^2 in a direction 53° South of West.
(D) The bug B moves with an acceleration 1 cm/s^2 in South-West direction.
8. As shown in figure, a rod of mass 'm' and length 'R' fixed on ring of same mass and radius R. The system is placed on perfectly rough surface and is gently displaced so that ring rotates. Find out the velocity of center of ring when rod becomes horizontal.



(A) $\sqrt{\frac{1}{2}gR}$ (B) $\sqrt{\frac{3}{2}gR}$
(C) $\sqrt{\frac{5}{2}gR}$ (D) $\sqrt{\frac{3}{10}gR}$

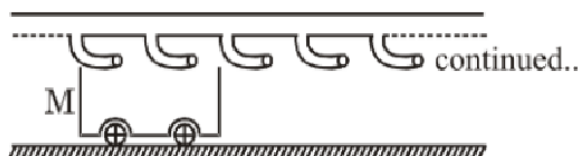
9. Estimate the minimum frequency of a γ -ray that causes a deuteron to disintegrate into a proton and a neutron. The masses of the particles are

$$m_d = 2.0141 m_u,$$

$$m_p = 1.0078 m_u,$$

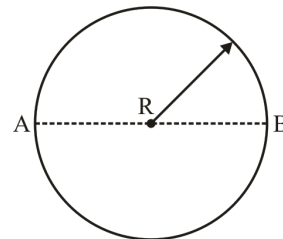
$$m_n = 1.0087 m_u$$

- (A) 5.4×10^{20} Hz
(B) 6.9×10^{11} Hz
(C) 6.2×10^6 Hz
(D) 7.3×10^{14} Hz
10. In a water supply system, horizontal nozzles are attached to the main supply pipe at regular interval of 10m. Each nozzle discharges water horizontally at the rate of 2 kg/s with velocity 5 m/s. An empty tank of mass 100 kg and length 20m is placed over smooth horizontal rails below the supply line initially at the rest. Find the velocity of the tank (in m/s) after 100sec. Ignore vertical gap between the tank and water supply line. Water started moving with the tank as it falls in it.



- (A) 2
(B) 4
(C) 6
(D) 8

11. A solid disk of mass M and radius R is pivoted about a horizontal axis through a point A on its rim as shown in the figure. It is released from rest in the shown position. What is the magnitude of the acceleration of point B when AB make an angle of 37° with the horizontal?



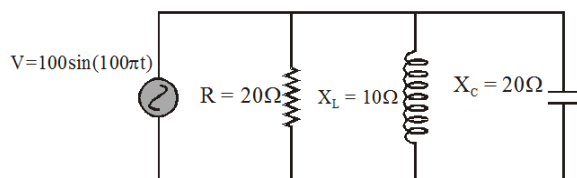
- (A) $\frac{4\sqrt{13}g}{15}$
(B) $\frac{8\sqrt{13}g}{15}$
(C) g
(D) $\frac{\sqrt{89}}{5}g$
12. A sample of radioactive material can decay by two processes A and B with half-lives $\frac{1}{2}$ hour and $\frac{1}{4}$ hour respectively. Now, the radioactive sample decays for 7 hour in such a way that for 1 hour it decays according to process A and for next hour decays according to B and this decaying with different processes in alternate hours continues upto 6 hour and then, in 7th hour, it decays simultaneously with both the processes. If N_0 nuclei of sample are present at $t = 0$, then number of radioactive nuclei after 7 hour would be

- (A) $\frac{N_0}{2^{16}}$ (B) $\frac{N_0}{2^{24}}$
(C) $\frac{N_0}{2^8}$ (D) $\frac{N_0}{2^5}$

13. 24 identical cells, each of internal resistance 0.5Ω , are arranged in a parallel combination of n rows, each row containing m cells in series. The combination is connected across a resistor of 3Ω . In order to send maximum current through the resistor, we should have

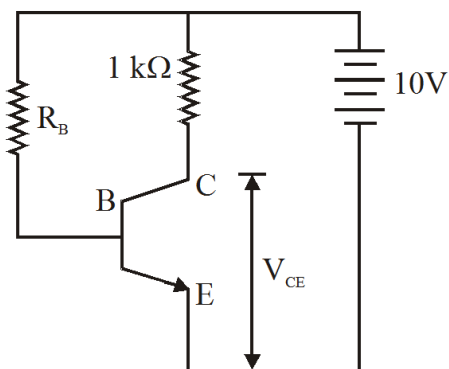
(A) $m = 12, n = 2$
 (B) $m = 8, n = 3$
 (C) $m = 2, n = 12$
 (D) $m = 3, n = 8$

14. In the given circuit, the rms value of current drawn from the source



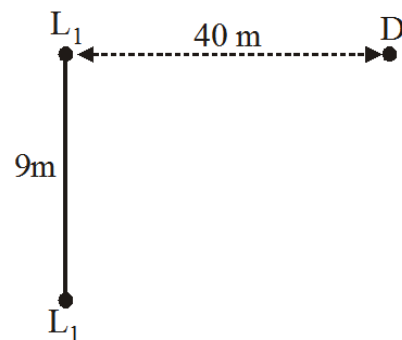
(A) 20 A
 (B) 10 A
 (C) 5 A
 (D) $5\sqrt{2}$ A

15. In the circuit shown here the transistor has a current gain $\beta = 100$. What should be the bias resistor R_B so that $V_{CE} = 5$ V ? (neglect V_{BE})



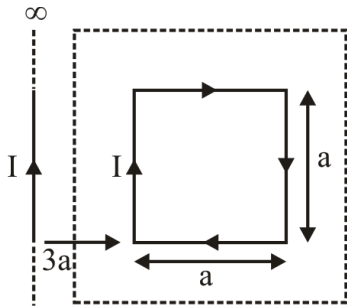
(A) $2 \times 10^3 \Omega$
 (B) $200 \times 10^3 \Omega$
 (C) $1 \times 10^6 \Omega$
 (D) 500Ω

16. Two loudspeakers L_1 and L_2 driven by a common oscillator and amplifier, are arranged as shown. The frequency of the oscillator is gradually increased from zero and the detector at D records a series of maxima and minima. If the speed of sound is 330 ms^{-1} , then the frequency at which the first maximum is observed is



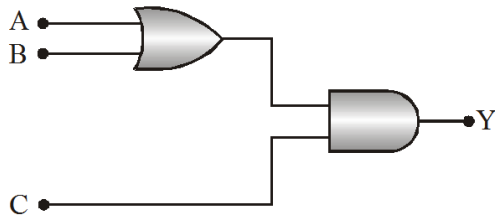
- (A) 165 Hz
 (B) 330 Hz
 (C) 495 Hz
 (D) 660 Hz
17. In an ideal heat pump, heat from inside at 277 K is transferred to a room at 300 K. The amount of heat (in joule) which will be delivered to the room for each joule of electrical energy consumed is
- (A) 12 J
 (B) 1 J
 (C) 25 J
 (D) 13 J

18. The work done in increasing the side length of square from a to $2a$ uniformly, as shown in figure.



- (A) $\frac{-\mu_0 I^2 a}{2\pi} \ln\left(\frac{15}{8}\right)$
 (B) $\frac{-\mu_0 I^2 a}{\pi} \ln\left(\frac{75}{16}\right)$
 (C) $\frac{-\mu_0 I^2 a}{2\pi} \ln\left(\frac{75}{16}\right)$
 (D) $\frac{-\mu_0 I^2 a}{\pi} \ln\left(\frac{15}{8}\right)$

19. To get output $Y = 1$ from circuit of figure, the inputs must be



- (A)

A	B	C
0	1	0

 (B)

A	B	C
1	0	0

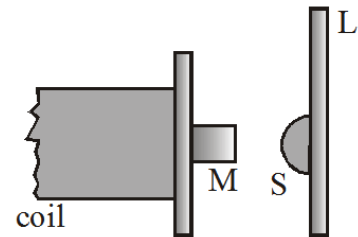
 (C)

A	B	C
1	0	1

 (D)

A	B	C
1	1	0

20. The diagram shows part of a magnetic relay, M is part of the core of the magnet L is part of the armature which is attached to the core when a current flows through the coil. S is a stud which stops the armature being attracted too strongly. Which line of the table gives the best materials for M, L and S ?



- (A)

M	L	S
Iron	Iron	Iron

 (B)

M	L	S
Iron	Iron	Copper

 (C)

M	L	S
Iron	Copper	Copper

 (D)

M	L	S
Copper	Copper	Copper

SECTION-II : (Maximum Marks: 20)

This section contains 10 questions Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is an Integer Value Type questions.

For each question, enter the correct integer value (In case of non-integer value, the answer should be rounded off to the nearest Integer).

Answer to each question will be evaluated according to the following marking scheme:

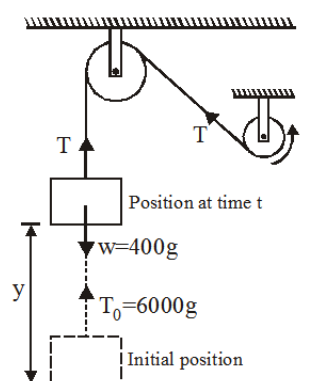
Full Marks : +4 If correct answer is selected.

Zero Marks : 0 If none of the option is selected.

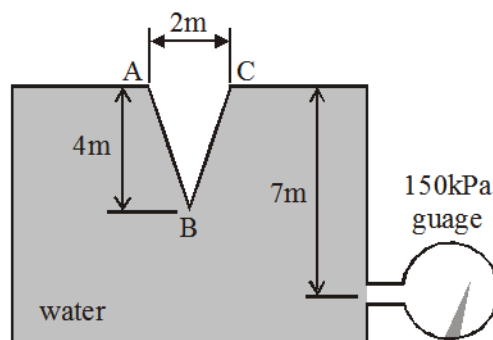
Negative Marks : -1 If wrong option is selected.

1. A ball is projected vertically upwards with a velocity 40 m/s returns back on the ground with velocity 30 m/s. If the air resistance is proportional to the speed of the ball. The airtime of the ball (in sec) is. Acceleration of free fall is 10 m/s^2 .

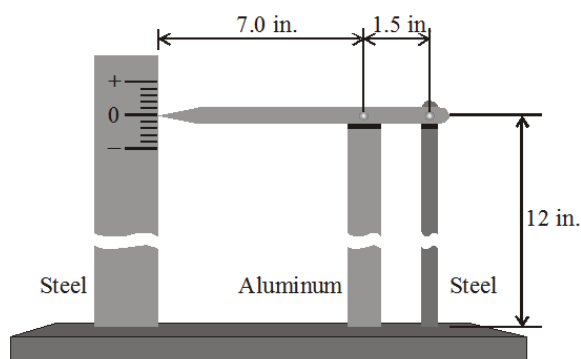
2. A body of mass 400 kg is suspended at the lower end of a light vertical chain and is being pulled up vertically (see figure). Initially the body is at rest and the pull on the chain is $6000g \text{ N}$. The pull gets smaller uniformly at the rate of $360g \text{ N}$ per each meter through which the body is raised. What is the speed (in m/s) of the body when it has been raised 5m ?



3. A container has water filled upto a height 7 m. A pressure guage at the bottom of the container measures the pressure as 150kPa. The net hydrostatic force (in kN) on the conical surface ABC as shown in the diagram.



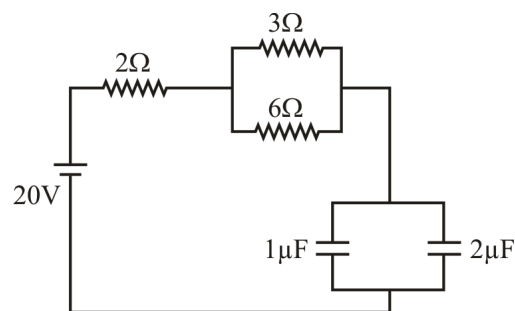
4. The movement of the pointer of figure with respect to the scale zero in response to a temperature increase of 60°F is $(x \times 10^{-3})$ inch. The value of x is : (Coefficient of thermal expansion are $6.6 \times 10^{-6}/^{\circ}\text{F}$ for the steel and $12.5 \times 10^{-6}/^{\circ}\text{F}$ for the aluminum.)



5. Three identical containers, 1, 2 and 3, are filled with equal quantities of ice at 0°C and placed outside. Identical heating elements are placed inside each container. The elements are powered by different voltages: $V_1 = 380 \text{ V}$, $V_2 = 220 \text{ V}$, and $V_3 = 180 \text{ V}$. In container 1, all ice melted in $t_1 = 4 \text{ min}$. In container 2, all ice melted in $t_2 = 20 \text{ min}$. How long (in minute) would it take to melt all the ice in container 3? Assume that the resistance of the heating elements remains constant and that the temperature of each container is constant throughout its volume at any given instant. Also assume the rate of heat loss to the surrounding is same for all three containers.

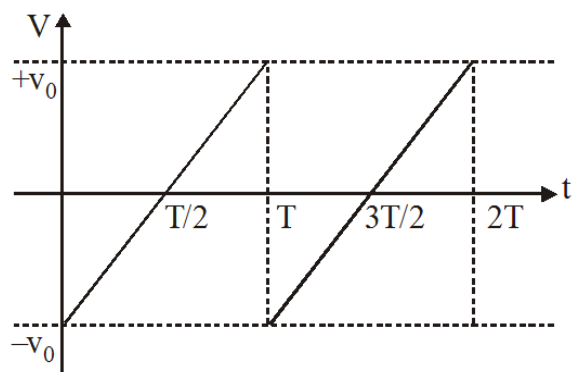
6. Bananas contain potassium, in the form of ^{39}K (93.258%), ^{40}K (0.0117%), and ^{41}K (6.7302%). The isotope ^{40}K is radioactive, with a half-life of $1.28 \times 10^9 \text{ yr}$. The total amount of potassium in a banana is 0.50 g. The activity of a banana (in Bq) is. (Take : $\ln 2 = 0.7$)

7. In the circuit shown the capacitors are initially uncharged. In a certain time the capacitor of capacitance $2 \mu\text{F}$ gets a charge of $20 \mu\text{C}$. In that time interval find the heat produced in 3Ω resistance in micro joule.



8. A vehicle approaches at night from a large distance. Only resolving a single source, at first you think it is a motorcycle. But as it gets closer, you can distinguish to headlights. If the headlights are separated by 1.3 m and the pupil of your eye has a diameter of 7.0 mm, how far away (in km) was the vehicle when the headlights became distinguishable ? Assume $\lambda = 580 \text{ nm}$.

9. The RMS value of the sawtooth voltage as shown in the diagram is. (Take : $v_0 = 8\sqrt{3}V$)



10. Two men A and B of mass M and $2M$ are standing 120° apart on a circular platform of mass $4M$ and radius R . Now men interchange their position with respect to disc by walking along periphery only. If $\Delta\theta$ represents magnitude of angle rotated by platform with respect to ground. Find $\frac{\Delta\theta_{\max}}{\Delta\theta_{\min}}$ (Platform is free to rotate about its centre, while men are standing at periphery)

SECTION-I : (Maximum Marks: 80)

This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

Full Marks : +4 If correct answer is selected.

Zero Marks : 0 If none of the option is selected.

Negative Marks : -1 If wrong option is selected.

- Oxidation

$$\text{As}_2\text{S}_3 \longrightarrow \text{H}_3\text{AsO}_4 + \text{H}_2\text{SO}_4$$

50% of the electrons released by 5 mole of As_2S_3 is used to reduce 35 moles of ions of element A as :

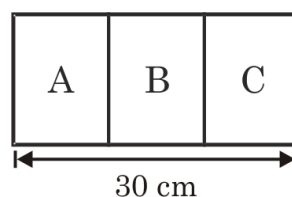
$$\text{A}^{+x} \longrightarrow \text{A}^{-x}$$

Find value of 'x' ?

(A) 1
(B) -1
(C) 2
(D) -2
- In 90 g of water, 2 moles of non-volatile solute B is added which slowly trimerises into an insoluble solid B_3 . This trimerisation process has rate constant $k = 6.93 \times 10^{-2} \text{ min}^{-1}$. Find vapour pressure of solution after 10 min of addition of solute 'B'. (V.P. of H_2O at $25^\circ\text{C} = 60 \text{ mm Hg}$)

(A) 55 mm Hg
(B) 45 mm Hg
(C) 50 mm Hg
(D) 52.5 mm Hg

- A rectangular box is fitted with two massless and frictionless very thin pistons as shown in the figure. Initially 1 mole of an ideal gas is present in each compartment at 1 atm and 300K. Now compartments A, B and C are maintained at 400K, 500K and 600K respectively. What will be the difference in initial and final separation between the two pistons assuming no heat exchange occurs between the compartments ?



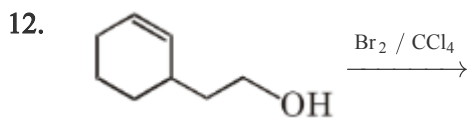
- (A) 2
(B) 4
(C) 6
(D) 0
- Consider the following equilibrium at 25°C

$$\text{CuSO}_4 \cdot 5\text{H}_2\text{O}_{(s)} \rightleftharpoons \text{CuSO}_4 \cdot 3\text{H}_2\text{O}_{(s)} + 2\text{H}_2\text{O}_{(g)}$$

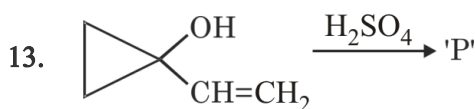
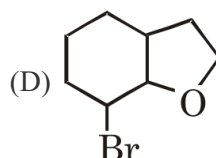
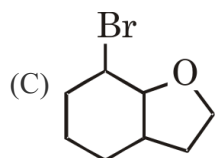
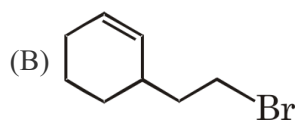
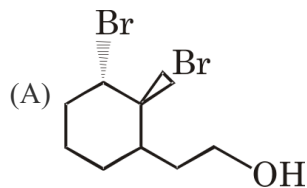
If this reaction takes place at two cities A and B having relative humidity 60% and 80% respectively. The vapour pressure of water at 25°C is 20 mm Hg and K_p for the above reaction is $4 \times 10^{-2} (\text{atm})^2$ at 25°C then $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ acts as efflorescent in which city?

(A) At A only
(B) At B only
(C) At both A and B
(D) Neither at A nor at B

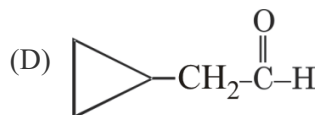
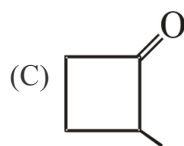
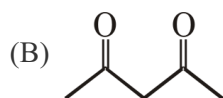
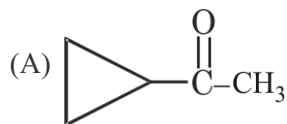
5. Which of the following is True?
- (A) The dilute solution of sodium in liquid ammonia does not conduct electricity
- (B) The dilute solution of sodium in aqueous ammonia is blue coloured solution
- (C) The different concentrations of Na^+ and K^+ are required across cell walls towards functioning of signal
- (D) Anhydrous MgCl_2 can not be used for extracting Mg using electrolytic method
6. Which of the following will not release NO_2 upon reaction with HNO_3 (in respective conditions).
- (A) Cu in dilute HNO_3 acid
- (B) Cu in conc. HNO_3 acid
- (C) Sn in conc. HNO_3 acid
- (D) Zn in conc. HNO_3 acid
7. Which of the following ligand can exhibit linkage isomerism when present in complex compound.
- (I) SO_4^{-2} (II) PO_4^{-3} (III) SO_3^{-2}
- (IV) CO_3^{-2} (V) CN^- (VI) SCN^-
- (VII) $\text{C}_2\text{O}_4^{-2}$
- (A) I, II, V, VI
- (B) V, VI
- (C) III, V, VI
- (D) V, VI, VII
8. Which of the following is incorrect?
- (A) One of the hydrolysis product of SF_4 is pyramidal tetravalent compound
- (B) SF_6 does not hydrolyse due to steric crowding
- (C) One of the hydrolysis product of NCl_3 will react with K_2HgI_4 in KOH
- (D) One of the hydrolysis product of AsCl_3 is pentavalent in major form
9. V_2O_5 reacts with alkalies to give
- (A) VO_4^{-3}
- (B) VO_4^+
- (C) VO^{+2}
- (D) VO_2^{+2}
10. Which of the following is not a green house gas?
- (A) CO_2
- (B) CH_4
- (C) O_3
- (D) N_2
11. Which of the following statements is incorrect?
- (A) ClO_2 is powerful oxidising agent
- (B) ClO_2 is diamagnetic in nature
- (C) ClO_2 undergoes disproportionation in basic medium
- (D) ClO_2 has higher bond angle compared to Cl_2O .



Major product is -

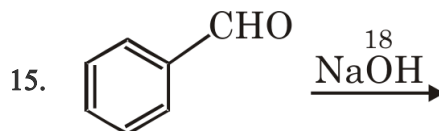


Identify 'P' in the reaction?

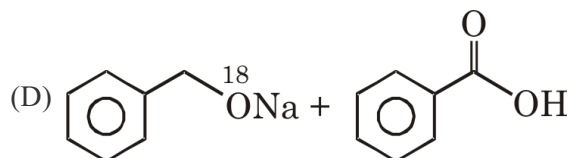
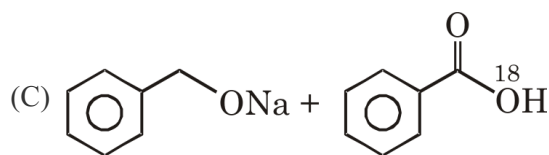
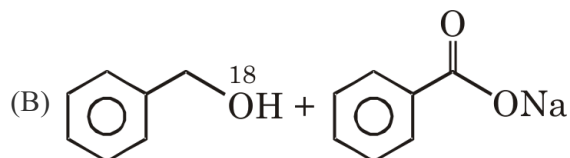
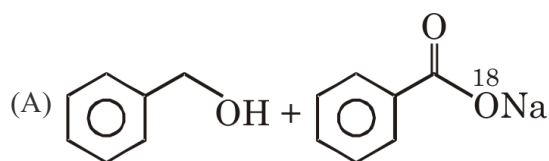


14. A new carbon-carbon bond formation is possible in -

- (A) Cannizzaro reaction
(B) Friedel-Crafts alkylation
(C) Clemmensen reduction
(D) None of these

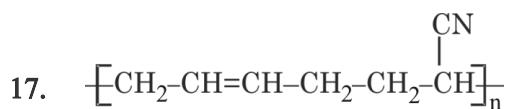


The products of the reaction are -



16. Morphine changes into Heroin by replacing

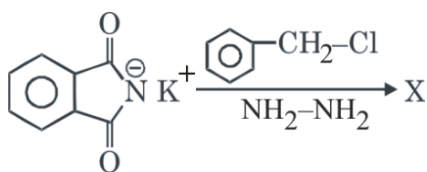
- (A) all -OH by -OCH₃
(B) one -OH by -OCH₃
(C) all -OH by -OAc
(D) one -OH by -OCH₃ and one -OH



The name of polymer is :

- (A) Polystyrene
(B) BuNa-S
(C) BuNa-N
(D) BuNa-CN

18. Gabriel phthalimide synthesis give a product amine 'X'.



X give test -

- (A) $\text{X} \xrightarrow{\text{Br}_2-\text{H}_2\text{O}}$ Brown colour disappear
(B) X Sweet smelling comp.
(C) $\text{X} \xrightarrow{\text{NaNO}_2+\text{HCl}}$ Yellow-Oily layer
(D) $\text{X} \xrightarrow{\text{CS}_2+\text{HgCl}_2}$ Musturd oil smell
19. In some F.C.C. crystal lattice, three types of atoms A, B and C are present. 'A' atoms occupy FCC lattice points, 'B' atoms occupy octahedral voids and 'C' atoms occupy tetrahedral voids. If all atoms along two body diagonals are removed and new empirical formula of the lattice is given as $\text{A}_x\text{B}_y\text{C}_z$, then find $x + y + z$ is.
- (A) 15
(B) 16
(C) 21
(D) 19

20. An aqueous solution of weak base (BOH) was titrated with a strong acid solution. The equivalence point was reached when 72.24 mL of 0.1 N strong acid solution was added. Also after adding 36.12 mL of same 0.1 N strong acid solution pH was found to be 9.2. Find value of pK_b of weak base (BOH) ?

- (A) 4.8
(B) 5.8
(C) 6.8
(D) 9.8

SECTION-II : (Maximum Marks: 20)

This section contains 10 questions Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is an Integer Value Type questions.

For each question, enter the correct integer value (In case of non-integer value, the answer should be rounded off to the nearest Integer).

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If correct answer is selected.

Zero Marks : 0 If none of the option is selected.

Negative Marks : -1 If wrong option is selected.

1. $\text{B}_2\text{H}_{6(\text{g})} + 6\text{Cl}_{2(\text{g})} \rightarrow 2\text{BCl}_{3(\text{g})} + 6\text{HCl}_{(\text{g})} \quad \Delta_r H = -1326 \text{ kJ}$
- $\text{BCl}_{3(\text{g})} + 3\text{H}_2\text{O}_{(\text{l})} \rightarrow \text{H}_3\text{BO}_{3(\text{g})} + 3\text{HCl}_{(\text{g})} \quad \Delta_r H = -112.5 \text{ kJ}$
- $\text{B}_2\text{H}_{6(\text{g})} + 6\text{H}_2\text{O}_{(\text{l})} \rightarrow 2\text{H}_3\text{BO}_{3(\text{s})} + 6\text{H}_{2(\text{g})} \quad \Delta H = -493.4 \text{ kJ}$
- $\frac{1}{2}\text{H}_{2(\text{g})} + \frac{1}{2}\text{Cl}_{2(\text{g})} \rightarrow \text{HCl}_{(\text{g})} \quad \Delta H = -92.3 \text{ kJ}$

From the above data, calculate molar enthalpy of sublimation of H_3BO_3 at 300 K (in kJ/mole).

2. The standard reduction potential for Cu^{+2}/Cu electrode is 0.34 V. The solubility product of $\text{Cu}(\text{OH})_2$ is 1.0×10^{-19} . The pH of solution at which the reduction potential for the electrode becomes 0.31 V if Cu^{+2} concentration is maintained by $\text{Cu}(\text{OH})_2$ salt ? $\left(\frac{2.303RT}{F} = 0.06 \right)$

3. $\text{H}_n\text{A} \rightleftharpoons \text{H}^{\oplus} + \text{H}_{n-1}\text{A}^{\ominus}$



If degree of ionization for successive deprotonation becomes half of its previous value, then find the value of 'n' if $\frac{[\text{H}^{\oplus}]}{[\text{A}^{n-}]} = 31$.

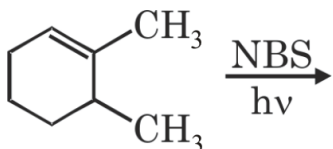
[Basicity of H_nA is 'n'].

4. How many of the following statements is/are correct?
- (a) Soap acts as emulsifier in its cleansing action.
- (b) Bredig's Arc method is particularly suitable for preparing sulphur sol.
- (c) Coagulation of sol can be caused by electrophoresis.
- (d) Greater is the zeta potential greater is stability of the colloid.
- (e) Methylene blue sol is a negatively charged sol whereas TiO_2 sol is a positively charged sol.
- (f) Lower the gold number, more is the protective power.
- (g) Lower the coagulation value, higher the coagulation power.
5. Find the value of $x + y$ where x and y indicate number of H atoms in $\text{H}_x\text{Fe}(\text{CO})_4$ and $\text{H}_y\text{Re}(\text{CO})_5$ respectively (Assuming both complex follow Sidgwick rule) (Atomic number : Re = 75)
6. How many of the following relations is/are correct?
- (a) $\text{Mg} > \text{Al}$ (electropositive character)
- (b) $\text{Al} > \text{Ga}$ (electronegativity)
- (c) $\text{N}^+ > \text{N}$ (ionization energy)
- (d) $\text{Se} > \text{S}$ (magnitude of ΔH_{eg})
- (e) $\text{He}^+ > \text{H}$ (size)
- (f) $\text{O}_2^{2-} < \text{O}_2$ (paramagnetic nature)
- (g) $\text{C}_2 > \text{N}_2$ (Number of π bonds)
- (h) $\text{F}_2 > \text{B}_2$ (bond order)
- (i) $\text{B}_2 > \text{F}_2$ (bond length)
- (j) $\text{NO}_2^- > \text{NO}_3^-$ (bond angle)

7. Amongst the following, the total number of compounds whose aqueous solution gives white precipitate with Pb^{2+} ions is

KI , NH_3 , Na_2CO_3 , K_2CrO_4 , NaCl , Na_2SO_3 ,
 Na_2S , KNO_3 , CH_3COONa

8. Number of possible product formed



9. $\text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{OH} \xrightarrow[\text{(2) CH}_2\text{N}_2(1 \text{ eq.})]{\text{(1) SOCl}_2} \text{Product.}$
 (3) AgOH , Δ
 (4) H_2O
 (5) H^+

$[\text{R} = \text{C}_{(n-2)}\text{H}_{2n-3}]$

If number of hydrogens in the product is 24.

The value of 'n' is :

10. Equanil is used in controlling depression and hypertension. The formula of Equanil is $\text{C}_x\text{H}_y\text{O}_z\text{N}_w$. Find $[2x+1] - \left(\frac{y+z-2w}{2}\right)$.

SECTION-I : (Maximum Marks: 80)

This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

Full Marks : +4 If correct answer is selected.

Zero Marks : 0 If none of the option is selected.

Negative Marks : -1 If wrong option is selected.

1. Let $S = \left[-\pi, \frac{\pi}{2}\right) - \left\{-\frac{\pi}{2}, -\frac{\pi}{4}, -\frac{3\pi}{4}, \frac{\pi}{4}\right\}$.

Then the number of elements in the set

$A = \left\{\theta \in S : \tan \theta (1 + \sqrt{3} \tan 2\theta) = \sqrt{3} - \tan 2\theta\right\}$ is

(A) 3

(B) 4

(C) 5

(D) 7

2. If α and β are the roots of equation

$x^2 - (5 + 3\sqrt{\log_3 5} - 5\sqrt{\log_5 3})x + 3(3^{(\log_3 5)^{\frac{1}{3}}} - 5^{(\log_5 3)^{\frac{2}{3}}} - 1) = 0$,

then the equation whose roots are $\frac{1}{\alpha}$ and $\frac{1}{\beta}$ is

(A) $x^2 - 5x - 3 = 0$

(B) $3x^2 + 5x - 1 = 0$

(C) $3x^2 - 10x + 2 = 0$

(D) $x^2 - 10x - 6 = 0$

3. For $t \in (0, 2\pi)$, if ABC is an equilateral triangle

with vertices A(sint, -cost), B(cost, sint) and

C(a, b) such that its orthocentre lies on a circle

with centre $\left(\frac{2}{3}, \frac{1}{3}\right)$, then $(a^2 + b^2)$ is equal to

(A) 8 (B) 5 (C) $\frac{8}{3}$ (D) $\frac{80}{9}$

4. A circle C_1 passes through the origin O and has diameter 4 on the positive x-axis. The line $y = 2x$ gives a chord OA of a circle C_1 . Let C_2 be the circle with OA as a diameter. If the tangent to C_2 at the point A meets the x-axis at P and y-axis at Q, then AP : PQ is equal to

(A) 1 : 5

(B) 2 : 5

(C) 3 : 5

(D) 4 : 5

5. Let the function $f(x) = 2x^2 - \log_e x$, $x > 0$, be decreasing in $(0, a)$ and increasing in $(a, 4)$. A tangent to the parabola $y^2 = 4ax$ at a point P on it passes through the point $(8a, 8a - 1)$ but doesn't pass through the point $\left(-\frac{1}{a}, 0\right)$. If the normal at P meets the curve again at Q, then slope of tangent at Q is

(A) $-9/2$

(B) $-2/9$

(C) $2/9$

(D) None of these

6. If $\frac{1}{2} < x < \frac{1}{\sqrt{2}}$ and $\frac{\sin^{-1} x}{\alpha} = \frac{\cos^{-1} x}{\beta}$, then the value of $\sin\left(\frac{\alpha\pi}{\alpha + \beta}\right)$ is

(A) $-2x\sqrt{1-x^2}$

(B) $2x\sqrt{1-x^2}$

(C) $4x\sqrt{1-x^2}$

(D) $-4x\sqrt{1+x^2}$

7. If $\lim_{x \rightarrow \infty} (8x^3 + mx^2)^{1/3} - nx$ exists and is equal to 1, then the value of $\frac{m}{n}$ is
- (A) $\frac{1}{6}$ (B) 6
(C) 3 (D) $\frac{1}{3}$
8. Let f and g be differentiable function satisfying $g'(a) = 2$, $g(a) = b$ and $f \circ g = I$ (where I is Identity function). Then $f'(b)$ is equal to
- (A) $1/2$ (B) 2
(C) $2/3$ (D) $3/4$
9. Let the unit vectors $\vec{a}, \vec{b}, \vec{c}$ be the position vectors of the vertices of a triangle ABC. If $\vec{F}$ is the position vector of the mid point of the line segment joining its orthocentre and centroid then $(\vec{a} - \vec{F})^2 + (\vec{b} - \vec{F})^2 + (\vec{c} - \vec{F})^2 =$
- (A) 1
(B) 2
(C) 3
(D) None of these
10. A box contains 12 mangoes out of which 5 are rotten and rest are good. Two mangoes are randomly taken out together. If it is known that atleast one of them is good, then the probability that both are good is :
- (A) $\frac{5}{8}$ (B) $\frac{3}{8}$
(C) $\frac{2}{5}$ (D) $\frac{4}{7}$
11. Let v and w be distinct, randomly chosen roots (real or complex) of the equation $z^9 - 1 = 0$. The probability that $1 \leq |v + w|$ is
- (A) $\frac{{}^6C_2}{{}^9C_2}$
(B) $\frac{3}{{}^9C_2}$
(C) $\frac{4}{9}$
(D) None of these
12. Calculate the standard deviation for the following :
- | | | | | | | | |
|----------------|---|---|---|----|----|----|----|
| Size of Team : | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
| Frequency : | 3 | 6 | 9 | 13 | 8 | 5 | 4 |
- (A) 1.607
(B) 4.507
(C) 7.709
(D) 2.335
13. The direction cosines ℓ, m, n of two lines are given by $p\ell + qm + rn = 0$ and $a\ell^2 + bm^2 + cn^2 = 0$ where p, q, r, a, b, c , are arbitrary constants. If these two lines are parallel then
- (A) $\frac{b+c}{p^2} + \frac{c+a}{q^2} + \frac{a+b}{r^2} = 0$
(B) $\frac{a^2}{p} + \frac{b^2}{q} + \frac{c^2}{r} = 0$
(C) $\frac{p^2}{a} + \frac{q^2}{b} + \frac{r^2}{c} = 0$
(D) $p^2(b+c) + q^2(a+c) + r^2(a+b) = 0$

14. For the hyperbolas $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ and $-\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ their eccentricities are x and $f(x)$ then $\int_2^4 \underbrace{f(x) \cdot f(x) \cdot \dots \cdot f(x)}_{10 \text{ times}} dx$ is

(A) 4 (B) $2\sqrt{2}$ (C) 6 (D) 8

15. Four Ellipse E_1, E_2, E_3, E_4 are drawn such that E_n touch E_{n-1} at end points of its major axis and E_n has foci at end points of minor axis of E_{n-1} where $n \in \{2, 3, 4\}$. If eccentricity of E_1 is $\frac{\sqrt{3}}{2}$ then eccentricity of E_4 is

(A) $\frac{1}{\sqrt{5}}$

(B) $\frac{2}{3}$

(C) $\frac{3}{\sqrt{23}}$

(D) $\sqrt{\frac{5}{14}}$

16. The smaller area bounded by the circle $x^2 + y^2 = 2$ and $y = f(x)$ above X-axis where $f(x)$ is a polynomial of least degree satisfying

$$\lim_{x \rightarrow 0} \left(1 + \frac{f(x)}{x^3}\right)^{1/x} = e \text{ is}$$

(A) $\frac{\pi}{2} - \frac{3}{5}$

(B) $\frac{\pi}{2} + \frac{3}{5}$

(C) $\frac{\pi}{2} + \frac{8}{5}$

(D) $\frac{\pi}{2} - \frac{8}{5}$

17. $\alpha, \beta, \gamma, \delta$ are in H.P. α, γ are roots of $t_1 x^2 - 4x + 1 = 0$ and β, δ are roots of $t_2 x^2 - 6x + 1 = 0$ then product of slopes of normals drawn to hyperbola $xy = c^2$ at $\left(ct_1, \frac{c}{t_1}\right)$ and $\left(ct_2, \frac{c}{t_2}\right)$ is

(A) 243

(B) 625

(C) 576

(D) None of these

18. The statement $[(p \wedge q) \vee (\sim p \wedge q) \vee (r \wedge \sim q)] \vee [p \wedge (\sim p \vee \sim q) \wedge q]$ is equivalent to

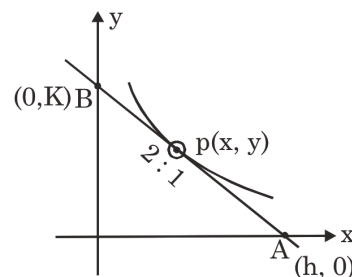
(A) $p \wedge q$

(B) $q \vee r$

(C) $p \vee q$

(D) $q \wedge r$

19. A tangent drawn to the curve $y = f(x)$ at $p(x, y)$ cuts x axis and y axis at A and B respectively such that $BP : AP = 2 : 1$, given that $f(1) = 1$.



(A) Equation of curve is $x \frac{dy}{dx} + 3y = 0$

(B) Equation of curve is $x + \frac{dy}{dx} - 2y = 0$

(C) Curve passes through (2, 2)

(D) Normal at (1, 1) is $2y = x + 1$

20. If the roots of equation $x^3 - 12x^2 + kx - 28 = 0$ are in A.P. then value of k is
- (A) 13
(B) 39
(C) 4
(D) -39

SECTION-II : (Maximum Marks: 20)

This section contains 10 questions Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is an Integer Value Type questions.

For each question, enter the correct integer value (In case of non-integer value, the answer should be rounded off to the nearest Integer).

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If correct answer is selected.

Zero Marks : 0 If none of the option is selected.

Negative Marks : -1 If wrong option is selected.

- $2(\cos x + \cos 2x) + (1 + 2\cos x) \sin 2x = 2\sin x$
Total no. of solutions in $[-\pi, \pi]$ is
- $p(x)$ is a cubic polynomial satisfying $p(-1) = 10$ and $p(1) = 6$, $p(x)$ maximised at $x = -1$ and $p'(x)$ minimum at $x = 1$, then $|10f(3)|$ is
- $(t-1)x + (3t+1)y + 2t + z = 0$
 $(t-1)x + (4t-2)y + (t+3)z = 0$
 $2x + (3t+1)y + 3(t-1)z = 0$
has non trivial for t_1 and t_2 and in this context ratio of $x : y : z = a : b : c$, when t has the smallest value, then $a + b + c +$ (sum of all the different value of t) is

- The exhaustive set of values of parameters m for which function $f(x) = 9x^3 + 9(m-1)x^2 + 3(m+5)x + 7$ is increasing for $\forall x \geq 0$ is $[k, \infty]$ than value of $6k^2$ is
- A and B are 2×2 matrices with $\text{tr}(A) = \text{tr}(B)$ and $\det(A - B) \neq 0$, where $\text{tr}(A)$ denotes trace of matrix A . Then the absolute value of $\frac{\text{tr}((A - B)^2)}{\det(A - B)}$ is
- Two circles of radii 2 and 1 touch each other externally. The intercept made by the transverse common tangent between two direct tangents is $a\sqrt{a}$. Then a is
- The number of natural numbers n such that the highest power of 6 in $n!$ is 117, is
- Number of ways $N = 68600$ can be expressed as product of 3 distinct co-prime integers is:
- Consider $f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ such that $f^2(x) + f^2(y) - 2xy = f^2(x+y) + 2(x(f(x) - f(x+y)) + y(f(y) - f(x+y)))$ where $f(1) = 2$ and $f(x) > x \forall x \in \mathbb{R}^+$ find $f(9)$.
- $$f(x) = \begin{cases} \left[\frac{x^2 - 4x + 3}{x^2 - 3x + 2} \right], & x < 1 \\ k, & x = 1 \\ x^2 + ax + b, & x > 1 \end{cases}$$
 where $[x]$ denotes greatest integer function. If $f(x)$ is differentiable at $x = 1$, $a^2 + b^2 + k^2 = ?$

Space for Rough Work

